

Advanced Programming

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SESSION 5 — 6 OCTOBER 2026



Would you notice?

Today's session

	MIN	WHAT
A	20	25 days of parking
B	20	You had one job · 28 of 28
C	25	Your spike
D	15	Homework 5



25 days of parking

What do you pay?

DAILY = 1000 # Ft

MONTHLY = 20000 # Ft, any 30 days in a row

```
def best_price(days):
```

You park for **25 days**. What do you pay?

25 days: two ways to pay

DAILY TICKETS

25 × 1,000

25,000 Ft

MONTHLY PASS

1 pass = any 30 days in a row

20,000 Ft

Two tests, one program

```
def best_price(days):  
    return days * DAILY           # the code: forgets the pass  
  
def test_by_the_agent():  
    assert best_price(25) == 25000 # copied from running the code  
  
def test_by_the_room():  
    assert best_price(25) == 20000 # said before running
```

The agent's test passes — it copied the code. **The room's test fails** — and now you know the code is wrong.

And 45 days?

The obvious fix:

```
def best_price(days):  
    return min(days * DAILY, MONTHLY)
```

Your 25-day test is now green. **What do you pay for 45 days?**

45 days: the room's answer

1 pass (30 days) + 15 daily tickets = **35,000 Ft.** The fix said **20,000.**

```
def test_45_days():  
    assert best_price(45) == 35000      # said before running  
  
def best_price(days):  
    passes, rest = divmod(days, 30)  
    return passes * MONTHLY + min(rest * DAILY, MONTHLY)
```

Two answers from outside caught two bugs. Now both tests are green.



You had one job

You had one job



A good hob. A good ring.

Budapest, Nyugati tér, 1980



Built from both ends. In the middle, **the two halves did not meet** in height.

The legend: two different sea levels.

The record: the work was rushed — the elements were tensioned before the glue had set. Each half looked fine on its own.

You had one job



Two good handles.

Mars Climate Orbiter, 1999

What happened. One team's software sent thruster data in **pound-force seconds**. The navigation software expected **newton seconds**. Every number was off by a factor of 4.45.

Why nobody noticed. Each program was right by its own rules. The data flowed for months; small course errors were noticed, but nobody checked them against an outside number.

The outside answer. Where the spacecraft actually was. It flew too low and was lost. A \$327 million mission.

Source: NASA Mars Climate Orbiter Mishap Investigation Board, Phase I Report, 10 Nov 1999.

You had one job



A good door. Good steps.

Laufenburg, 2003



Photo: Taxiarchos228, Wikimedia Commons (CC BY-SA 3.0). Sources: Kanton Aargau press release; swissinfo.ch, "Meereshöhe ist nicht gleich Meereshöhe".

A bridge over the Rhine, Germany to Switzerland, built from both banks.

Germany measures height from the **North Sea**, Switzerland from the **Mediterranean**: 27 cm apart.

The gap: **54 cm**. Caught on site, December 2003.

You had one job



Good steps. A good fence.

Ariane 5, Flight 501, 1996

What happened. 37 seconds after launch the rocket destroyed itself. A velocity value no longer fit into a 16-bit number.

Why nobody noticed. The code came from Ariane 4 and had worked for years. It was **never tested with Ariane 5's own flight path** — a faster rocket.

The outside answer. Ariane 5's real trajectory data. The inquiry asked for exactly that test.

Source: Ariane 501 Inquiry Board Report (Lions report), 19 July 1996.



28 of 28

Would your tests notice?

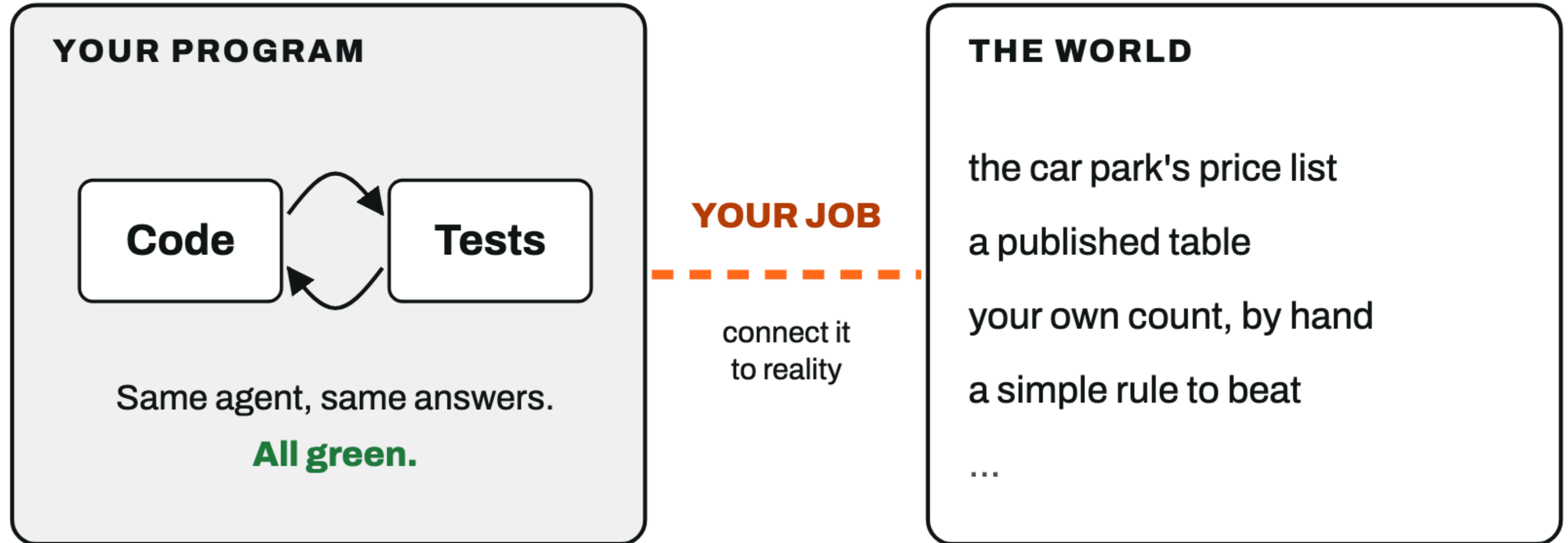
Your agents wrote **20,000 lines of tests**. They all pass.

We changed **one line** in each project, so that its main README promise became false. Then we ran your tests.

28 of 28: nothing went red.

Not a verdict on you. All 28 miss the same thing as the parking code: **a link to the real world**.

Your program agrees with itself



You have one job





Your spike

Your spike

In Homework 4 you already measured your program against something real.

- 1 Find your project on the next slides.
- 2 Finish the sentence: **A test would go red if ...**

That sentence is where Homework 5 starts.

AlterEgo Finder

What it does. Give it a face photo; it shows the three most similar faces in a collection and how sure it is.

What you found in Homework 4. With 500 faces in the collection, 99 of 100 strangers got a match it called very sure.

So right now: in a large collection, “very sure” does not mean the person is there.

A test would go red if ...

Blocky

What it does. It blocks distracting websites on a weekly schedule; if you open one anyway, you get a page with better things to do.

What you found in Homework 4. Of ten sites chosen in advance, one (YouTube) still got through in your usual browser; you then changed the extension to step in earlier.

So right now: nobody has checked that the browser and the app together really show the “blocked” page.

A test would go red if ...

Budget Planner

What it does. You enter your income; it suggests a budget in six categories and warns when you overspend.

What you found in Homework 4. Sorting a real month of your own spending, about a quarter of the money fell under “other”.

So right now: a small month fits; a full month with cash is untried.

A test would go red if ...

CAUSAL LAB

What it does. You give it data on a treated and an untreated group; it plots both, says whether they moved alike beforehand, and estimates the treatment's effect.

What you found in Homework 4. On simulated data with nothing wrong in it, the app's trust score averaged 57.6 out of 100, because several checks warn for nothing.

So right now: a low score does not yet tell users whether their data really has a problem.

A test would go red if ...

Central European air traffic

What it does. It records planes over Central Europe on a schedule, so traffic over time can be asked about later.

What you found in Homework 4. You counted the fully captured hours over two weeks: only 23 of 337.

So right now: Answers about traffic over time would rest on data that is mostly gaps.

A test would go red if ...

ChatGPT Ads Industry Dashboard

What it does. It groups ChatGPT ad accounts by industry and compares their spending, clicks and sales between industries.

What you found in Homework 4. The gap between industries looked bigger than the one real campaign's week-to-week swing: 143% versus 56%. But only one industry has real data; the others are example figures.

So right now: the industry differences on the chart are not yet real.

A test would go red if ...

Data Jobs & Skill-Gap Finder for Early-Career Professionals

What it does. For newcomers: from European data job ads, it shows which skills each data job asks for most.

What you found in Homework 4. You checked its skills against your own reading of 50 ads: right almost 99 times in 100.

So right now: an ad that says “R&D” could be counted as asking for the skill “R”, and nothing would notice.

A test would go red if ...

deriva

What it does. It compares two rounds of a labour survey and reports what changed, keeping real changes apart from cosmetic ones.

What you found in Homework 4. You tried three rounds it had never seen: 3 of 3 loaded unchanged.

So right now: it reads new rounds, but still calls many cosmetic changes real.

A test would go red if ...

Detective game with LLM-driven NPCs

What it does. A murder mystery: you question three suspects played by an AI, then name the killer.

What you found in Homework 4. In 30 real conversations, every reply passed the game's check that suspects stick to what they know.

So right now: the short intro the AI adds before an answer is not checked, so a secret could slip out there.

A test would go red if ...

European bank financial data loader

What it does. Pick a European bank and year; it gathers its key figures from two public sources into one table.

What you found in Homework 4. Checked against the bank's own reports, the figures agree, and its risk ratio comes out at 44.5%.

So right now: right for one bank and one year; others are unchecked.

A test would go red if ...

Explainable Fraud Detection

What it does. It reads a list of card payments and picks the ones worth a closer look, saying why.

What you found in Homework 4. On a public dataset where fraud is already marked, 68 of its 500 picks were really fraud.

So right now: most of the short list is not fraud.

A test would go red if ...

Financial Planner

What it does. From your bank statements, it shows where your money went, by category, month by month and over the year.

What you found in Homework 4. You sorted a fresh month by hand: shops it knew were almost always right, new shops only about two in three.

So right now: spending at unfamiliar shops often lands in the wrong category.

A test would go red if ...

Market Research Platform

What it does. Researchers load survey results and get a report organised by research goal.

What you found in Homework 4. You checked whether the summary tables can show answers by age group: they cannot — each estimate could be anything from zero up.

So right now: the screen is right to warn that such breakdowns are not available.

A test would go red if ...

MarketLens

What it does. Beginners practise investing with play money: it says whether stocks, bonds or gold will rise or fall, explains why, and tracks the portfolio at real prices.

What you found in Homework 4. Replaying past days, its forecasts were right slightly less often than a rule that always says “up”: 52 versus 55 in 100.

So right now: nobody has checked that an old forecast is scored against the right day’s price.

A test would go red if ...

Menstrual Cycle Phase Calculator and Targeted Nutrition Planner

What it does. From your last period and cycle length, it tells you your cycle phase, which nutrients to focus on, and writes a grocery list.

What you found in Homework 4. You looked for medical sources behind its advice: 7 of 12 pairings have one, but mostly for the whole cycle, not one phase.

So right now: users are not told how strong the evidence behind each suggestion is.

A test would go red if ...

Ms. Match

What it does. You photograph your own clothes; it reads today's forecast, and one click puts together an outfit shown on a photo of you.

What you found in Homework 4. Adding ten of your own photos, you measured that the browser holds about 63 of your own garments, close to the original guess.

So right now: on a rainy day the outfit button ignores the weather, so it may still pick trainers or high heels.

A test would go red if ...

Photo Duplicate Finder

What it does. It finds copies and near-identical shots in a photo folder and suggests which to keep. It never deletes anything.

What you found in Homework 4. On about 300 of your own photos, only the strictest settings kept groups clean; the default chained unrelated photos together.

So right now: nobody knows yet whether today's setting keeps bursts together and merely similar shots apart.

A test would go red if ...

PixWars

What it does. A team game over a local network: build bridges between floating platforms and break the other team's bed.

What you found in Homework 4. On your home wifi, between a laptop and a phone, no game messages were lost, but the slowest ones were too slow for smooth play.

So right now: nobody knows yet whether two laptops can play on the classroom wifi.

A test would go red if ...

plant_care

What it does. Each houseplant gets a card that counts down to its next watering.

What you found in Homework 4. You looked up 12 plants with a gardening society: none gives a number of days, only soil dryness or season.

So right now: The countdown is a starting guess, and the app now says so.

A test would go red if ...

Process Swimlane Generator

What it does. From a table describing a workplace process, it draws a diagram: a lane per department, a box per task, arrows for the order.

What you found in Homework 4. On five real, anonymised processes, four worked; one failed because it really has a yes-or-no branch.

So right now: nobody has checked that long processes stay readable, or that each arrow lands on the next task.

A test would go red if ...

RC Atlas

What it does. A programme manager, a field coordinator and a monitoring officer pass tasks along: assign, do, submit, check, approve, and the project status follows.

What you found in Homework 4. On a fresh install, a fixed ten-step workflow saved every step: 10 of 10.

So right now: nobody has checked that each person sees only the buttons meant for their role.

A test would go red if ...

RepoScout

What it does. Type a topic; it ranks GitHub projects and says how alive the field is.

What you found in Homework 4. You compared GitHub's "last updated" date with each project's real last change: 5 of 50 landed in the wrong group.

So right now: A few projects it calls active may not really have changed lately.

A test would go red if ...

Route Tariff and Capacity Fee Calculator

What it does. A gas trader picks a pipeline point and period and gets the cheapest way to book capacity.

What you found in Homework 4. Booking a little more time than needed was cheaper in four of ten realistic cases.

So right now: it now books extra time, but never considers a whole-year booking.

A test would go red if ...

sentmine

What it does. From an English text you just read, it makes a flashcard for each new word, in its real sentence.

What you found in Homework 4. You checked every card from one newspaper article by hand: 4 of 103 were really new words.

So right now: Most cards are known words or names, so the learner still sorts them by hand.

A test would go red if ...

Stardew Valley Auto-Gifter

What it does. A Stardew Valley add-on: it plans tomorrow's gifts for villagers from a chest, you approve the plan, and it hands them over as you walk past.

What you found in Homework 4. The specially marked chest kept its colour at all 3 openings, also after saving and reloading.

So right now: a villager who already got a gift today may still get another one.

A test would go red if ...

SurveyGuard: Numeric Careless Responding Detector

What it does. It marks survey respondents who seem to answer carelessly, for example the same answer everywhere, and says why.

What you found in Homework 4. On real public survey data, one of its checks, at the old setting, flagged about 7 in 10 ordinary respondents.

So right now: someone who gives the same answer to every question is not marked suspicious.

A test would go red if ...

Toto Estimator

What it does. Enter this week's Toto matches; it tips home win, draw or away win from past results.

What you found in Homework 4. On a real English season it had not seen: 40% right, against 46% for always tipping a home win.

So right now: Always tipping a home win currently scores better than its tips.

A test would go red if ...

Volleyball Statistics Tracker

What it does. It turns European volleyball match reports into team and player summaries.

What you found in Homework 4. You compared its team totals with the federation's own sheet: 22 of 24 matched exactly, the rest only by rounding.

So right now: Correct today, but only a small hand-made sample is checked, not a real full report.

A test would go red if ...

Homework 5 — Make it usable

- Use your program **once more for real** this week, on your own input. Expected before, got after — in the log.
- The first mismatch becomes **a test that can fail**: show it red, then green.
- If it can't be used yet: remove **the smallest thing** that stops you, then use it.
- Branch `hw5-usable`, pull request, merge commit — as before.

Due Saturday 11 October, 20:00.